Home Automation 1.0 – Project Documentation

Nov 3, 2026 · @Samuel

Overview

Home Automation 1.0 turns the Sony A95K into a morning screen that wakes with Samuel's alarm. It is built on Home Assistant, with as little new hardware as possible.

Goal 1 for the 1.0 release:

- A new landing page for the TV.
- The alarm switches the TV on and shows today's weather and schedule. The schedule comes from Apple Calendar and Outlook (a different account), and Samuel chooses which calendars appear.
- Control from the smartphone through a web-hosted backend on a Raspberry Pi.
- One overview of the last-watched TV shows and anime, with a flag when a new episode is out. Launching the app from the overview comes later.

Working rules:

- Check existing solutions (Home Assistant) before writing custom code.
- Review a rough UI sketch before anything is built.
- Ask instead of assuming whenever something is open.
- Record every decision, so the project doubles as a software-engineering learning exercise.

This document is the single source of truth. Requirements, decisions and open questions are updated here as the project moves.

Hardware and network

All hardware is owned now; the Raspberry Pi 5 (8 GB) was bought on 3 Nov 2026.

Device	Role in v1.0
Sony OLED XR-55A95K (Google TV)	Shows the landing page and the morning screen; Home Assistant switches it on. A Google account is signed in

Device	Role in v1.0
Sonos Arc + Sonos Sub (Gen3)	Soundbar and subwoofer; alarm speakers. System on S2
Sonos One SL (2×)	Alarm speaker
Apple devices (iPhone)	Phone control through the Home Assistant app; Apple Calendar source; location for leaving home
Raspberry Pi 5, 8 GB (bought), official 27 W supply, Flirc passive case	Runs Home Assistant OS, the hub of the system
SanDisk Extreme Portable SSD 1 TB (SDSSDE61-1T00, owned)	Boot and data drive for the Pi, connected to a blue USB 3 port. It is wiped when Home Assistant OS is installed

Network facts:

- Samuel built the network himself. All devices are in one /24 network (subnet mask 255.255.255.0, confirmed by a Fing scan on 3 Nov 2026). Samuel set static IPs on his devices.
- The ISP router gives no admin access: no port forwarding, no DHCP reservations, no router-level DNS.
- The ISP's fibre router is also the DHCP server. Its network can't be changed without admin access.
- **Warning about a /16 split by the third octet:** a device with an address outside the router's subnet could still talk to other devices that also use /16, but it would lose internet access. The router sends the replies for that address to the internet instead of back into the home. Phones and other devices that get their address automatically (DHCP, /24) couldn't reach it either.
- The network already runs as /24, so the /16 plan was only the old setup. Recommendation: stay in the router's /24 subnet and group devices by ranges in the last octet, outside the router's DHCP range (ADR-007).
- **DHCP overlap risk:** the phone and the PC probably got their addresses from the router, right between the static devices. The router's DHCP range likely overlaps the static IPs, which can cause address conflicts. Ask the ISP's support for the exact range.
- The Pi can use a 10 Gbit wall outlet. A Raspberry Pi 5 has a 1 Gbit port, far more than this workload needs.

Devices in the network

The full device list with addresses is kept in a private note in the Claude project, not here, because this document is exported to GitHub. For v1.0 the relevant devices are the Sony TV, the four Sonos speakers (Arc, Sub, two One SL), the iPhone and the router.

Requirements

Nineteen requirements make up v1.0, all decided on 3 Nov 2026; remaining details are under Open questions.

ID	Requirement	Status
R1	New landing page on the TV, shown at alarm time and on demand	Decided
R2	When an alarm rings, the TV turns on and shows the morning screen: weather and the schedule	Decided
R3	Calendars: Apple Calendar (iCloud), an Outlook work account and an Outlook school account	Decided
R4	A management web app (phone and computer) to pick which calendars are included; all picked calendars merge into one calendar view	Decided
R5	Small schedule preview for today and tomorrow; a per-event switch in the management web app decides whether an event appears in it; new events are shown by default, and the switch on the first date of a recurring event applies to the whole series	Decided
R6	Several independent alarms, managed in Home Assistant	Decided
R7	Each alarm runs on single weekdays or a custom group of weekdays	Decided
R8	No snooze. A ringing alarm stops with the Sonos speaker's button if possible, otherwise from the phone, otherwise by itself after 15 minutes	Decided
R9	Waking the TV is an on/off option per alarm	Decided

ID	Requirement	Status
R10	Each alarm plays on a chosen Sonos zone (room or group) from Spotify; for now Spotify's Discover Weekly playlist; fixed volume, today level 5, no ramp	Decided
R11	After an alarm the TV shows the kiosk page for at most 15 minutes; it switches off earlier when Samuel leaves home (phone location)	Decided
R12	Alarms skip the public holidays of the canton of St. Gallen	Decided
R13	A good phone UI to create and edit alarms; control from the smartphone through the web backend	Decided
R14	Watch overview of last-watched shows and anime: Crunchyroll, Netflix, Disney+, Prime Video, Apple TV+, HBO Max; an hourly refresh is enough	Decided
R15	The new-episode flag means the episode is available to Samuel on his service	Decided
R16	External access through cloud services at low cost; single user (Samuel only)	Decided
R17	The management web app shows a pop-up when Home Assistant, the OS or an add-on has an update, with its version and release notes; nothing installs until Samuel confirms it there. Nothing installs at night (21:00 until 2 hours after the last alarm): an update confirmed in that time is queued and runs 2 hours after the last alarm of the next day (at 10:00 on a day without an alarm); one confirmed during the day installs right away, and the result is shown in the management web app and sent to the phone. Pending updates are announced once a day at 21:00 with a phone notification	Decided
R18	Backup alarm on Sonos: if the Pi fails, a Sonos alarm stored on the speaker rings 10 minutes after the planned alarm (Spotify, or the Sonos chime if Spotify can't be reached)	Decided

ID	Requirement	Status
R19	Watchdog: the Pi checks in with a free outside monitoring service every few minutes; if the check-ins stop, the phone gets a notification	Decided

Backlog for later versions:

- Launch the streaming app from the watch overview.
- 2.0: skip the alarm on days with an Outlook event marked as holiday.
- 2.0: organise the calendars and remove duplicates.
- 2.0: replace the TV's software entirely. As far as known, Sony allows no other TV operating system, so 2.0 starts with Projectivy Launcher as the new home screen (ADR-015).

Decision log

Twenty-one decisions so far: nineteen accepted, one proposed and waiting for Samuel, one rejected. Each row is a short Architecture Decision Record (ADR); each can later move into its own file in the GitHub repository.

ID	Decision	Why	Status
ADR-001	Home Assistant OS on a Raspberry Pi is the platform	Official integrations already cover the Sony TV, Sonos, iCloud and weather; custom code only fills the gaps. Hardware: Raspberry Pi 5 8 GB, the own SanDisk SSD and a Flirc passive case, bought on 3 Nov 2026	Accepted
ADR-002	Home Assistant owns the alarms; Sonos only plays the sound	Sonos sends no event when an alarm rings (HA community), so the TV could not react reliably	Accepted
ADR-003	Cloud services by default for external access	Samuel's choice on 3 Nov 2026; low cost matters, see ADR-008	Accepted
ADR-004	Several alarms on custom weekday groups, no snooze; stop with the Sonos button, the phone or a timeout	Samuel's answers on 3 Nov 2026. The Sonos button works by Home Assistant noticing that the speaker was paused; to be tested	Accepted

ID	Decision	Why	Status
ADR-005	A small add-on on the Pi fetches Crunchyroll history every hour	Crunchyroll has no public history API (mal-sync); the account history also covers the TV. Samuel accepted the terms-of-service risk for this private use on 3 Nov 2026	Accepted
ADR-006	The TV shows dashboards through Fully Kiosk; test with the free version, buy the licence later	Home Assistant can tell it which page to show and bring it to the front (HA docs)	Accepted
ADR-007	The network stays on the router's /24 subnet; devices are grouped by ranges in the last octet	Without router access the router keeps its /24, so addresses outside it would lose internet access (see Hardware and network)	Proposed
ADR-008	Remote access through Tailscale (free) instead of Home Assistant Cloud	Samuel chose it on 3 Nov 2026: Home Assistant Cloud's EUR 75 a year is too steep, while Tailscale is free and keeps Home Assistant off the public internet (see Remote access options)	Accepted
ADR-009	This doc is the working copy; versions go into the GitHub repo as PDF; later switch to Markdown in the repo	Samuel's answer on 3 Nov 2026; this reading is to confirm	Accepted
ADR-010	All picked calendars merge into one calendar view, chosen on a settings page; duplicates are handled in 2.0	Samuel's answer on 3 Nov 2026; some calendars are imported and may duplicate events	Accepted
ADR-011	The TV switches off after 15 minutes at the latest, or earlier when Samuel leaves home, detected by the Home Assistant iPhone app	Samuel's answer on 3 Nov 2026. Needs remote access, because the phone reports leaving home over the internet	Accepted
ADR-012	Public holidays come from Home Assistant's Workday integration	Swiss holidays differ by canton; Samuel's canton is St. Gallen	Accepted

ID	Decision	Why	Status
ADR-013	The Spotify playlist is saved as a Sonos favourite, and Home Assistant starts that favourite	Starting a Sonos favourite is simpler and more reliable than controlling Spotify directly	Rejected
ADR-014	Home Assistant starts the Spotify playlist directly (Spotify integration) and plays it on the chosen Sonos zone	Replaces ADR-013, which Samuel rejected on 3 Nov 2026. Needs Spotify Premium, which Samuel has	Accepted
ADR-015	The TV keeps the normal Google TV home screen in 1.0; Projectivy Launcher is tested in 2.0	Samuel's choice on 3 Nov 2026. The morning screen runs in Fully Kiosk and needs no launcher; Projectivy replaces the home screen, which belongs to the 2.0 goal of an own TV interface. Projectivy is free, premium optional	Accepted
ADR-016	Calendars and the per-event preview switch are managed in one management web app, opened on the phone or computer; nothing is marked inside the calendar apps	Samuel's choice on 3 Nov 2026. Keeps work and school calendars untouched and puts all settings in one place	Accepted
ADR-017	The Pi sits in the bedroom in a passive, fanless case; backups run automatically at night	Samuel's choice on 3 Nov 2026: silence matters more than speed. Raspberry Pi calls cooling optional for normal use; heavy jobs may slow down a little, which is harmless at night	Accepted
ADR-018	Updates are never installed automatically; a pop-up in the management web app announces them and Samuel confirms each one by hand	Samuel's choice on 3 Nov 2026. Updates have broken the Sonos and TV integrations before, so release notes are read first. Home Assistant lists each pending update with its version and release notes and offers an install action the page can call	Accepted

ID	Decision	Why	Status
ADR-019	No updates at night: from 21:00 until 2 hours after the last alarm, a confirmed update is queued and installed 2 hours after the last alarm of the next day	Samuel's choice on 3 Nov 2026. An update that breaks something can never hit an alarm, and Samuel can check the result when he is home, with the evening left to fix it. Updates are announced daily at 21:00, when Samuel is usually home; the result of a queued update goes to the phone. Samuel's rule: the alarm is not optional, so nothing installs at night. On a day without an alarm, queued updates run at 10:00	Accepted
ADR-020	One Sonos alarm "Backup", created in the Sonos app and kept in sync by Home Assistant: every evening it is set to 10 minutes after the next alarm on the same zone and switched on; after the Home Assistant alarm is stopped, Home Assistant switches it off for that morning	Samuel's choice on 3 Nov 2026, instead of a phone backup. Sonos alarms are stored on the speaker and fall back to the chime when the music source fails (Sonos community). Fail-safe: the backup is on by default and only a working Home Assistant can switch it off. Home Assistant can change existing Sonos alarms but not create them; only one alarm is changed, once a day, because changing several at once is unreliable	Accepted
ADR-021	A watchdog: the Pi sends regular check-ins to a free outside monitoring service, which notifies the phone when they stop	Samuel's choice on 3 Nov 2026. Home Assistant cannot report its own failure, so an outside service has to notice the silence (dead man's switch). It also covers a Pi that is down for several days, when the backup alarm keeps its last time. The service is chosen during the build	Accepted

Remote access options

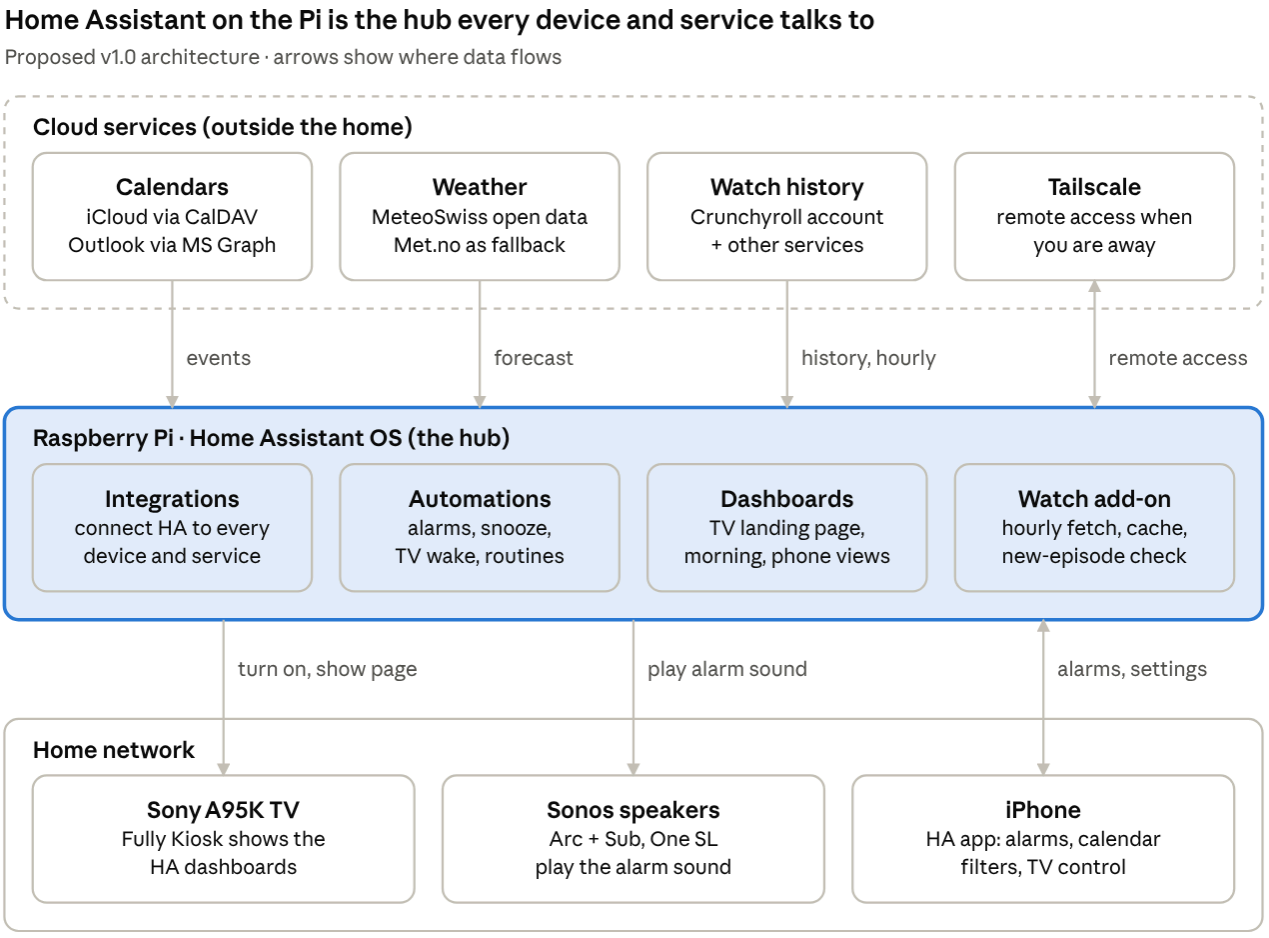
Tailscale is the chosen option (ADR-008). It's free, it needs no router access, and Home Assistant never becomes reachable from the public internet. Every option below uses an external service, as decided in ADR-003.

Option	Cost	What it needs	Exposure	Verdict
Tailscale	Free Personal plan: up to 6 users, unlimited devices	Tailscale add-on on Home Assistant, Tailscale app on the iPhone with the VPN connected	Private: only devices in your own network (tailnet)	Chosen
Cloudflare Tunnel	Tunnel free; own domain needed, about USD 10 a year (SmartHomeScene)	Community Cloudflared add-on, domain on Cloudflare, a trusted_proxies entry in the config	Public web address behind Cloudflare	Good if no VPN app on the phone is wanted
Home Assistant Cloud / Link	EUR 7.50 a month or EUR 75 a year	Only a sign-in	Public address through Nabu Casa, encrypted	Simplest, but the most expensive
ZeroTier	Free tier	ZeroTier network and app	Private	Works, smaller Home Assistant community
WireGuard, port forwarding, DuckDNS	Free	An open router port	Public port	Not possible: needs router access

One thing matters for R11 (TV switches off early when you leave home). The iPhone reports leaving home over the internet, so the Tailscale VPN must stay connected on the phone. Some users saw location updates fail over Tailscale until they set up the app again (HA community). Test this early.

Architecture

Everything runs through Home Assistant on the Pi; no device talks to another directly.



v1.0 architecture · 3 layers, 11 components

Home Assistant pulls calendars, weather and watch history from the cloud and drives the TV and the Sonos speakers on the home network. When Samuel is away, the phone reaches the Pi through Tailscale.

Risks and limitations

The biggest risks sit where the project depends on interfaces nobody promised to keep stable: Sonos, Google TV, Crunchyroll and the two Microsoft 365 accounts.

Risk	Effect	Mitigation
The Sonos local interface is unofficial; switching UPnP off or an HA update can break it	Alarm sound fails	Keep UPnP on, read release notes before updating, phone alarm as backup at first

Risk	Effect	Mitigation
Stopping the alarm with the Sonos button relies on Home Assistant seeing the speaker pause	The button may not stop the alarm reliably	Phone button and automatic timeout as fallbacks (R8)
Google TV does not officially allow a custom home screen	Launcher override can break after updates	Bring the kiosk page forward by automation, independent of the launcher
Launching apps on the TV is fragile; an HA update in Aug 2026 broke it once	TV opens the wrong app or none	One script for all TV commands, so a fix is needed in one place only
Waking the TV from deep standby needs "Remote start"	Higher standby power (accepted by Samuel)	Wake-on-LAN as fallback
Crunchyroll's API is undocumented and bot protection may block scripts	Watch overview loses Crunchyroll	Fetch only hourly; a real browser on the Pi as plan B
"Available to me" needs regional availability data that most services don't publish	New-episode flag wrong for some services	Crunchyroll's own episode list covers it; other services to research in build step 7
Work and school Microsoft 365 accounts may block third-party apps or calendar publishing	No Outlook events from that account	Ask both IT departments first; published calendar link as fallback
iCloud CalDAV only sees calendars stored in iCloud	Subscribed calendars are missing	Check on the settings page which calendars show up
Leaving-home detection needs the phone to reach Home Assistant from outside	TV doesn't switch off when leaving	Keep the VPN connected; the 15-minute limit still applies (R11)
Changing the subnet mask without router access	Devices outside the router's subnet lose internet	Stay in the router's subnet (ADR-007)
Community add-ons (HACS) can break or be abandoned	Features stop after an update	Pin versions, keep backups

Risk	Effect	Mitigation
The Pi is a single point of failure for the alarm	Alarm does not ring	Sonos backup alarm stored on the speaker (ADR-020), watchdog notification (ADR-021), reliable power supply and SSD, automatic restart

Open questions

Three questions are still open; ticked items are done. Answers given on 3 Nov 2026 are now in Requirements and the Decision log.

Project setup

- [x] ~~Documentation (ADR-009): this doc stays the working copy, each version goes into the repo as a PDF, and later we switch to Markdown in the repo. Is that right?~~

Alarms

- [x] ~~Check in the Sonos app that UPnP is on (Account → Privacy & Security → Connection Security → UPnP). Confirmed on 3 Nov 2026.~~
- [x] ~~Update lock (ADR-019): nothing installs from 21:00 until 2 hours after the last alarm. Decided on 3 Nov 2026: on a day without an alarm, queued updates run at 10:00.~~

Calendars

- [ ] Ask the work and school IT whether third-party calendar access (Microsoft Graph) or a published calendar link is allowed, and whether work calendar data may sit on a home server.
- [x] ~~Per-event preview switch (R5, ADR-016): decided on 3 Nov 2026. New events are shown by default, so none are forgotten; the switch on the first date of a recurring event applies to the whole series.~~

Watch overview

- [ ] Netflix, Disney+, Prime Video, Apple TV+, HBO Max: Trakt VIP (about USD 60 a year, tracks them automatically) or self-built fetchers per service (free, more fragile)? To research before build step 7.

Hardware and network

- [x] ~~Which hardware to buy: Raspberry Pi 5 (4 GB about CHF 110–142, 8 GB about CHF 160–179, toppreise.ch) plus power supply, case and SSD, or Home Assistant Green (CHF 179, ready to use)? Decided on 3 Nov 2026: Raspberry Pi 5 8 GB with the official 27 W supply and the Flirc passive case, bought at digitec; storage is the own SanDisk SSD.~~

- [ ] Network: the Fing scan confirmed /24 (the /16 plan and the split by level were only the old setup). Still open: the router's DHCP range (ask the ISP's support).
- [x] ~~TV launcher: Projectivy Launcher (one-time premium about EUR 7) or Google TV's free "Apps only mode"? Decided on 3 Nov 2026: Projectivy moves to 2.0 (ADR-015).~~

Build plan

The next step is the UI sketch. Nothing is built until Samuel has reviewed it.

1. Answer the remaining open questions, buy the hardware and extend the repository structure (`docs/` , `docs/adr/` , later `homeassistant/`).
2. UI sketch: TV landing page, morning screen and the phone alarm screen. Samuel reviews and changes it.
3. Foundation: install Home Assistant OS on the Pi, put the configuration under Git, set up remote access.
4. TV test: Sony control with a pre-shared key, "Remote start", Fully Kiosk. Test waking the TV from deep standby at real alarm time.
5. Data sources: iCloud calendar, Outlook calendar, MeteoSwiss weather.
6. Alarms in Home Assistant plus the phone alarm UI. Run them alongside the existing Sonos alarm for a week before switching over.
7. Watch overview: Crunchyroll add-on, the other services, the new-episode flag.
8. Launcher polish, final tests, release 1.0.

Setup guides

Free up storage on the Sony TV

Do this before installing Fully Kiosk. Uninstalling unused apps frees the most space; clearing the cache is safe; clearing data resets an app. Settings opens from your profile icon at the top right of the Google TV home screen.

1. Check how full the TV is: Settings → System → Storage → Internal shared storage. The exact path varies by model (Sony).
2. Uninstall apps you don't use: Settings → Apps → See all apps → the app → Uninstall. Preinstalled apps can often only be disabled.
3. Clear the cache of large apps: Settings → Apps → See all apps → the app → Clear cache. This deletes temporary files such as thumbnails; you stay logged in (Sony).
4. Only for an app that misbehaves: same menu → Clear data. This resets the app to a fresh install and logs you out.

5. Restart the TV and check the storage again.

Check the home network

Fing on the iPhone is the easiest tool. It's free and lists every device in the network with its IP address, manufacturer and MAC address (SourceForge).

1. On the iPhone: Settings → Wi-Fi → tap (i) next to your network. Note the IP address, subnet mask and router. A subnet mask of 255.255.255.0 means /24.
2. Install Fing from the App Store and run a network scan.
3. Compare the list: devices you set by hand (TV, Sonos) show your static IPs; phones and laptops got theirs from the router. The automatic ones show roughly which range the router hands out.
4. The exact DHCP range is only visible in the router's settings; the ISP's support can tell you.
5. For learning: on a laptop, `nmap -sn 192.168.x.0/24` does the same scan from the command line (replace x with your network).

Glossary

Terms used in this document, with the meaning they have in this project.

Term	Meaning
Add-on	A program packaged as a Docker container that runs next to Home Assistant on the Pi, e.g. the planned watch-history fetcher
ADR	Architecture Decision Record: a short note of one decision, its context and its consequences
API	Application Programming Interface: the defined way one program asks another for data or actions
Automation	A Home Assistant rule of the form trigger, condition, action, e.g. "at 06:30 on weekdays, wake the TV"
CalDAV	Open standard for reading calendars over the network; iCloud supports it
Dashboard	A Home Assistant web page made of cards; the TV and phone screens are dashboards

Term	Meaning
DHCP	The router service that hands out IP addresses automatically
Entity	One thing Home Assistant knows about, e.g. a calendar, a speaker or an alarm switch
HACS	Home Assistant Community Store: installs community-made integrations and cards
Home Assistant (HA)	Open-source home-automation platform; Home Assistant OS is its Linux distribution for the Pi
Integration	A Home Assistant module that connects to one device or service
Kiosk browser	A locked-down browser that shows one web page full screen and can be controlled remotely
Launcher	The app that draws the TV's home screen
OAuth	A standard login flow that gives an app limited access to an account without sharing the password
Presence detection	Knowing whether someone is at home, here from the iPhone's location in the Home Assistant app
Scrobbling	Recording automatically what you watch or listen to, as it plays
Single point of failure	One part whose failure stops the whole system, here the Pi for the alarm
Static IP	A fixed address set on the device itself instead of one handed out by DHCP
Subnet mask	Defines which addresses count as the local network; /24 means, for example, 192.168.1.0 to 192.168.1.255
Undocumented API	An interface a service uses internally but never promised to keep stable for others
UPnP	A local network protocol that Home Assistant uses to control Sonos

Term	Meaning
VPN	An encrypted tunnel that connects a device outside the home to the home network, e.g. Tailscale
Wake-on-LAN	A network message that switches a sleeping device on

Sources

- Home Assistant: Sony Bravia TV integration
- Home Assistant: Fully Kiosk Browser integration
- Home Assistant community: Sonos alarm trigger
- Nabu Casa: Home Assistant Link
- mal-sync: Crunchyroll history via website session
- Universal Trakt Scrobbler: supported services
- Raspberry Pi 5 product page
- Tailscale free plan (Costbench)
- Nabu Casa vs Tailscale vs WireGuard vs Cloudflare Tunnel (ESP32.co.uk)
- Home Assistant remote access methods compared (SmartHomeScene)
- Cloudflare Tunnel on Home Assistant (Pi My Life Up)
- HA community: Tailscale and location updates
- Sony: check storage on the TV
- Sony: clear app data and cache
- Raspberry Pi 5 prices (toppreise.ch)
- Home Assistant Green (Swiss-Domotique)